Dear moderator,

Please find the awarded criteria marks and comments for candidate **jhl859** below.

A/4	B/4	C/3	D/3	E/6	Total
4	4	2	3	6	19

Criterion A: Presentation
A very well-written IA. It is logically developed, has a nice flow, and is easy to follow. The exploration has a promising introduction with a clear rationale and a focused aim, which is fully met, a body, and a concluding section. The sections link in very well with each other. The student successfully explores and applies the mathematics, explaining her steps and reasoning at all times with no irrelevant or repetitive calculations.

Criterion B: Mathematical communication
Appropriate and consistent mathematical language is used at all times. The mathematical notation is, apart from forgetting to write "dx" at the bottom of page 16, error-free. All key terms are defined and explained very well. An appropriate number of equations, tables and supporting diagrams are used effectively.

Criterion C: Personal engagement
This is a very creative and original IA where the student voice can be heard throughout. The student incorporated a personal interest into the work, which drives the exploration forward. The IA contains unfamiliar mathematics that is self-taught and explained in a way such that a fellow student can comprehend. The student taught herself how to use the software "Geogebra" and used it well to produce supporting and insightful graphs.

Criterion D: Reflection
The student consistently reflects on her mathematical working and reasoning - critical reflection is seen at any stage really and it helps the IA develop and move forward. Implications of the results are discussed. The concluding section contains further critical reflection.

Criterion E: Use of mathematics
Relevant mathematics commensurate with the level of the course is used. Both familiar as well as unfamiliar mathematics is clear and error-free. The student uses an appropriate level of accuracy at all times and demonstrates sophistication and rigour by explaining all of her steps, showing and "talking through" her mathematical working at any stage.

Mathematics: analysis and approaches Higher Level
Internal Assessment 2023

Candidate: jhl859

Calculating Brazil's Gini coefficient

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Calculating Brazil’s Gini coefficient

1. Introduction, rationale and aim

Inequality is a global phenomenon that remains to be unsolved in a variety of fields, including education, resources, healthcare and wealth. The global distribution of wealth and income is very unjust, with countries in the European Union or North America being much wealthier and thus having greater opportunities to provide their citizens with essential services and resources than countries towards the southern hemisphere of our planet. But also on a smaller economic scale the distribution of income and wealth can be inequitable within a country. South Africa and Brazil for instance have been considered to have the highest inequality rating in income distribution in the world. One of my mother’s closest friends moved to Brazil one year ago, working in an international school in the country's largest city Sao Paulo, she has experienced the two extremes of the rich and poor in her everyday life, teaching among the offspring of the highest 20% of the population while encountering families begging for help in front of supermarkets. Telling us about her experiences deeply unsettled our family and motivated me to research further into the income inequality on our planet and individual countries, choosing Brazil as a focused real life example. When it came to determining a topic for my mathematical exploration, I wondered how I could apply my mathematical skills and knowledge to this problem and conducted a mathematical analysis of the country’s Lorenz curve and Gini coefficient, which can be used to determine the distribution of income and wealth.

Therefore the aim for this IA is to construct a Lorenz curve from cumulative data about Brazil's income distribution taken from the World Data Bank and then use three different mathematical methods to calculate the Gini coefficient. The final value will be compared to the actual published Gini coefficient of 0.49 in 2020. *(Our World in Data)* I will therefore first construct Brazil's Lorenz curve from a given set of data points and then use three mathematical approaches to determine the Gini coefficient from the curve. The three methods include the Trapezium sum method, simultaneous equations and Lagrange interpolation. Lastly, the results will be discussed, evaluated, and reflected upon.

2. The Lorenz curve and the Gini coefficient

The Gini coefficient is a tool in economics used to represent the income or wealth inequality within a population group, usually an economy. It can take up values of 0 to 1, where 0 represents perfect equality and 1 represents maximum inequality in distribution of income or wealth *(The Investopedia Team)*. It measures how much a group's income or wealth distribution deviates from the perfectly equal distribution, represented by the line of equality y = x on the Lorenz curve (figure 1). The Lorenz curve plots the cumulative percentages of total income earned in an economy on the y-axis against the cumulative number of receivers in the population of that economy from lowest to highest incomes on the x-axis. It begins with the poorest households or individuals referred to as the lowest 20% and continues up to the highest 20%. *(Hayes, Adam)*

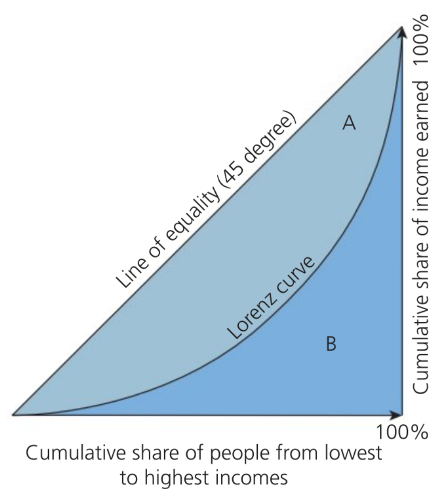

Figure 1: The Gini coefficient (Hoang, Paul)

To construct the Lorenz curve in order to later on determine the Gini coefficient from it, data about the cumulative population and cumulative income must be found. *(Ramzai, Juhi)*
The Gini coefficient can then be determined by finding the area between the Lorenz curve and line of absolute equality as a percentage of the total area under the line. This leads to the following formula (equation 1) to calculate the Gini coefficient, where A is the area between the line of equality and Lorenz curve and B is the area beneath the Lorenz curve, see figure 1 *(Hoang, Paul)*.

Equation 1: $Gini\ coefficient = \frac{A}{A+B}$

The Gini coefficient generally takes on a value of 0.24 to 0.63 and is not considered to be an absolute indicator of an economy's income but rather the dispersion of income in the economy's population. This can be interpreted to be fairly equal for values below 0.3, whereas a concerning income gap is indicated by a Gini coefficient over 0.4 expressing inequality of income distribution. *(CFI Team)* In order to calculate Brazil's Gini coefficient it is necessary to determine area A and B of Brazil's Lorenz curve to apply equation 1. The next section of this exploration will therefore focus on constructing Brazil's Lorenz curve.

3. Constructing Brazil's Lorenz curve

The real life example I will explore in this IA focusses around Brazil's income share data from 2020 which I first want to plot on a Lorenz curve in order to then determine the Gini coefficient mathematically from that Lorenz curve, thereby demonstrating the interconnectedness of economics and mathematics in the context of measuring economic inequality thus proofing the applicability of mathematical tools to determine economic indicators like the Gini coefficient. The following most recent data was taken from *The World Bank,* a financial institution part of the United nations. *(World Bank)*

3.1 Data - Brazil 2020

Income share held by lowest 20% → 4.5% of total income
Income share held by second 20% → 8.6% of total income
Income share held by third 20% → 12.9% of total income
Income share held by fourth 20% → 19.3% of total income
Income share held by highest 20% → 54.7% of total income

Table 1: Brazil's data on the inequality of distribution of income *(World Bank)*

Cumulative Population %	Cumulative Income %
0%	0.00%
20%	4.50%
40%	13.10%
60%	26.00%
80%	45.30%
100%	100.00&

These six given data points in table 1 will be used to determine the Gini coefficient from the Lorenz curve.

3.2 Processed data:

Plotting the data gives the following graph where the red Lorenz curve represents the income distribution in Brazil and the blue line is the perfect line of equality.

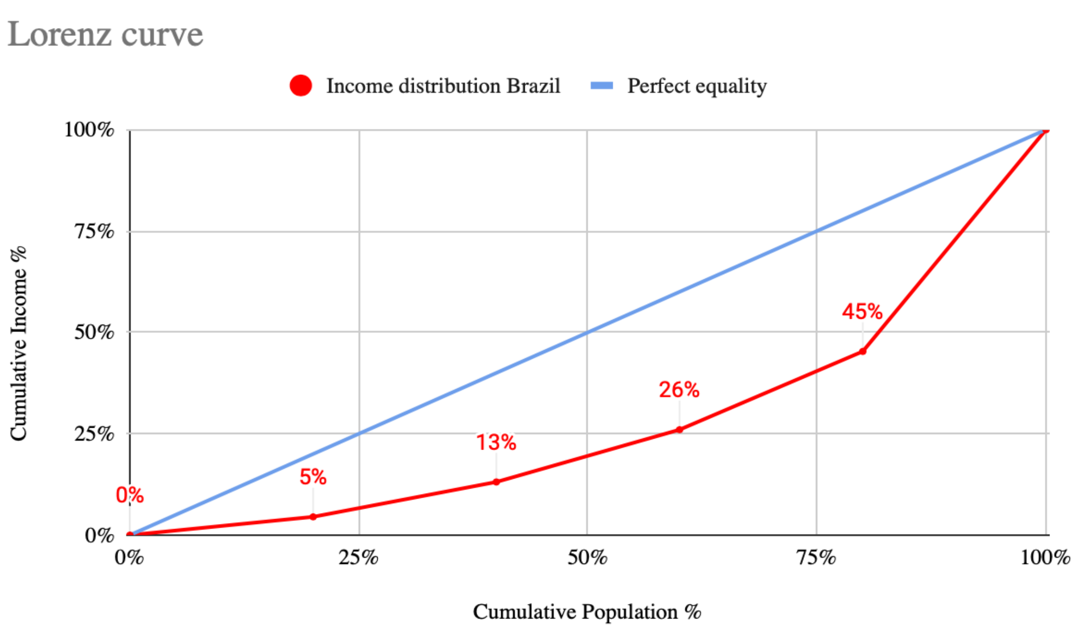

Figure 2: Brazil's Lorenz curve plotted from data using google sheets

In theory the red function should have a continuous curved shape *(Hoang, Paul)* which due to the limited data points used to construct the function has not been attained. These linear sections of the Lorenz curve will hence affect the accuracy of the area determined using the trapezium sum method. Therefore I will later approximate Brazil's Lorenz curve more accurately using simultaneous equations and interpolation methods to attain a continuous curved shape of the Lorenz curve which will allow for a more accurate calculation of the actual Gini coefficient.

4.1 Determining Brazil's Gini coefficient using the Trapezium area sum method

Riemann's sum method was our opening exercise when beginning the integration unit in our course. It refers to dividing the area beneath a curve into smaller rectangles in set intervals. By calculating every individual rectangles area, the sum of all rectangles combined approximates the area beneath the curve. Considering the shape of Brazil's constructed Lorenz curve, this method can be applied with the alternative use of trapezia. By calculating the individual trapezium area beneath the Lorenz curve for a set interval and summing these areas up, the total area *B* can be found using this approach, with the formula for each individual trapezium being:

Equation 2: $Area\ trapezium = \frac{1}{2} \times (a + b) \times h$

Since *a* and *b* denote the lengths of the parallel sides of a trapezium, they will take on a specific value of y, the cumulative income, whereas h represents the distance between the two parallel sides and will thus take on the cumulative population value, remaining constant at 0.2, the base length on along the x-axis.

For Brazil's Lorenz curve, 5 trapezia will be calculated as indicated in figure 3.

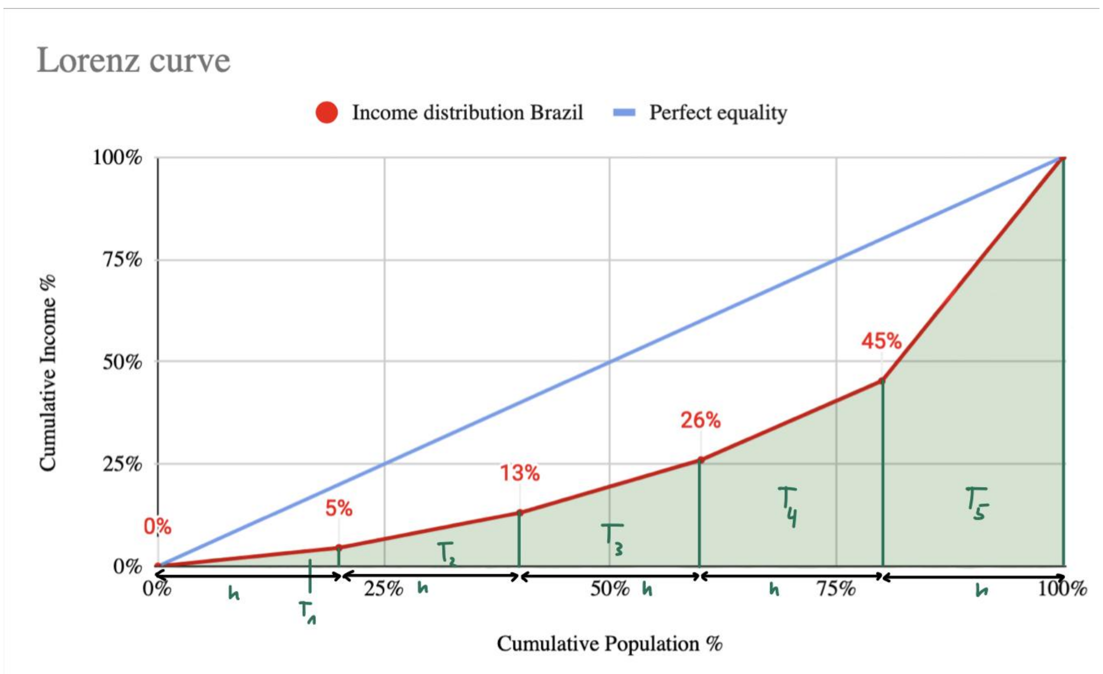

Figure 3: Brazil's Lorenz curve expressed in trapezia using google sheets

The trapezium formula can be alternatively expressed for a set interval beneath the Lorenz curve in this equation:

Equation 3: $Area\ under\ Lorenz\ curve\ = \frac{(I_N + I_{N+1})}{2} \times 0.2$

Where I represents the cumulative income taking on the previously introduced values of the trapeziums parallel sides a and b, as displayed in the following table.
It must be noted that for the first trapezium, I_N takes on the value 0, it may therefore be considered a triangle.

Table 2: Brazil's data for equation 4

N	Value of I_N	Value of h
1	0	0.2
2	0.045	0.2
3	0.131	0.2
4	0.26	0.2
5	0.453	0.2
6	1	0.2

From the values in the table the area beneath the Lorenz curve will be calculated by the formula:

Equation 4: $Area\ B\ = \sum_{N=1}^{6} \frac{I_N + I_{N+1}}{2} \times 0.2$

For demonstration purposes, Brazil's income distribution was substituted into the extended sigma notation.

$$\sum_{N=1}^{6} = \frac{0+0.045}{2} \times 0.2 + \frac{0.045+0.131}{2} \times 0.2 + \frac{0.131+0.26}{2} \times 0.2 + \frac{0.26+0.453}{2} \times 0.2 + \frac{0.453+1}{2} \times 0.2$$

$Area\ B\ = 0.2778\ units^2$

To find A, the area between the equality line and the Lorenz curve, the calculated value of B must be subtracted from the total area beneath the line of equality. The base for the triangle formula used to calculate the area is 1 since $P_{max} = 1.000$ and the height is 1 since $I_{max} = 1.000$. Hence, the total area beneath the line of equality can be calculated as follows:

$A_{equality} = \frac{1}{2} \times 1 \times 1$

$A_{equality} = 0.5\ units^2$

Now the Area A between the equality line and the Lorenz curve is

$A = 0.5 - 0.2778$

$A = 0.2222\, units^2$

Calculating the Gini coefficient using equation 1

$Gini\ coefficient = \frac{0.2222}{0.5}$

$Gini\ coefficient = 0.4444$

Rounded to two significant figures the Gini coefficient of 0.44 would indicate a 44% concentration of income distribution in Brazil which refers to a rather unequal distribution of income. It indicates that there is potential social or political tension in the country that results in a larger income gap and thus greater economic inequality. *(Hoang, Paul)*

As mentioned earlier, the Trapezium sum method is likely to be the least accurate because the Lorenz curve is constructed of only 6 data points and therefore only 5 trapezia were constructed to calculate the area. Therefore the following sections will aim to determine a more accurate equation of the Lorenz curve and integration of the area beneath.

4.2. Determining Brazil's Gini coefficient using simultaneous equations

In my research on how to integrate the Lorenz curve I needed to find a way to determine a more precise equation for the Lorenz curve which then can be integrated to find the area beneath. Finding the equation of a polynomial function using interpolation methods results in a more continuous curve and thus integrating gives a more accurate result of the area beneath, justifying the second approach explored to determine Brazil's Gini coefficient. Interpolating the given cumulative income and cumulative population data from the world data bank will allow us to find the equation of the polynomial function using a system of equations.

Assume the polynomial $P_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ is the equation we want to find for Brazil's Lorenz Curve such that it interpolates the n+1 points (x_0, y_0), (x_1, y_1), (x_n, y_n).

For Brazil's data the points would be:

Table 3: Brazil's data for simultaneous equations

n	x_n	y_n
0	0	0
1	0.2	0.045
2	0.4	0.131
3	0.6	0.26
4	0.8	0.453
5	1	1

To find the polynomial equation we are required to find the coefficients of the polynomial $P_n(x)$ denoted by *a,* hence we can write a system of equations for every single point that needs to be interpolated and converted into the matrix form. *(Shao, Jun) (The Math Guy)*

$$P_n(x_n) = y_n$$

$$a_0 + a_1x_n + a_2x_n^2 + \ \dots\ + a_{n-1}x_n^{n-1} + a_nx_n^n = y_n$$

Since we have six points given we will be approximating a 5th order polynomial where n = 5

$$P_5(0) = a_0 + a_1 \times 0 + a_2 \times 0^2 + a_3 \times 0^3 + a_4 \times 0^4 + a_5 \times 0^5 = 0$$

$$P_5(0.2) = a_0 + a_1 \times 0.2 + a_2 \times 0.2^2 + a_3 \times 0.2^3 + a_4 \times 0.2^4 + a_5 \times 0.2^5 = 0.045$$

$$P_5(0.4) = a_0 + a_1 \times 0.4 + a_2 \times 0.4^2 + a_3 \times 0.4^3 + a_4 \times 0.4^4 + a_5 \times 0.4^5 = 0.131$$

$$P_5(0.6) = a_0 + a_1 \times 0.6 + a_2 \times 0.6^2 + a_3 \times 0.6^3 + a_4 \times 0.6^4 + a_5 \times 0.6^5 = 0.26$$

$$P_5(0.8) = a_0 + a_1 \times 0.8 + a_2 \times 0.8^2 + a_3 \times 0.8^3 + a_4 \times 0.8^4 + a_5 \times 0.8^5 = 0.453$$

$$P_5(1) = a_0 + a_1 \times 1 + a_2 \times 1^2 + a_3 \times 1^3 + a_4 \times 1^4 + a_5 \times 1^5 = 1$$

In order to simplify the system of equations we evaluate the exponents and multiply by the coefficients *a.*

$$P_5(0) = a_0 + 0 + 0 + 0 + 0 + 0 = 0$$

$$P_5(0.2) = a_0 + 0.2a_1 + 0.04a_2 + 0.008a_3 + 0.0016a_4 + 0.00032a_5 = 0.045$$

$$P_5(0.4) = a_0 + 0.4a_1 + 0.16a_2 + 0.064a_3 + 0.0256a_4 + 0.01024a_5 = 0.131$$

$P_5(0.6) = a_0 + 0.6a_1 + 0.36a_2 + 0.216a_3 + 0.1296a_4 + 0.07776a_5 = 0.26$

$P_5(0.8) = a_0 + 0.8a_1 + 0.64a_2 + 0.512a_3 + 0.4096a_4 + 0.32768a_5 = 0.453$

$P_5(1) = a_0 + a_1 + a_2 + a_3 + a_4 + a_5 = 1$

We want to find 6 unknowns using a system of equations in order to determine the equation of P(x). Using the algebraic tool of the Vandermonde matrix in figure 4 by substituting the known values will allow us to determine the missing coefficients. *(Shao, Jun)*
Therefore the system of equations can be rewritten in matrix equation form, see figure 4, where the Vandermonde matrix is denoted by the set of x raised to each integer power up to the data points number minus one. It thus evaluates the polynomial P(x) for the given data set by solving for the coefficients in the system of equations.

$$\begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & x_1^4 & x_1^5 \\ 1 & x_2 & x_2^2 & x_2^3 & x_2^4 & x_2^5 \\ 1 & x_3 & x_3^2 & x_3^3 & x_3^4 & x_3^5 \\ 1 & x_4 & x_4^2 & x_4^3 & x_4^4 & x_4^5 \\ 1 & x_5 & x_5^2 & x_5^3 & x_5^4 & x_5^5 \\ 1 & x_6 & x_6^2 & x_6^3 & x_6^4 & x_6^5 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{bmatrix}$$

Figure 4: Matrix equation with Vandermonde matrix[1]

Now substituting the known data points from Brazil's Lorenz curve for the respective x and y values in the matrix, gives figure 5.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0.2 & 0.04 & 0.008 & 0.0016 & 0.00032 \\ 1 & 0.4 & 0.16 & 0.064 & 0.0256 & 0.01024 \\ 1 & 0.6 & 0.36 & 0.216 & 0.1296 & 0.07776 \\ 1 & 0.8 & 0.64 & 0.512 & 0.4096 & 0.32768 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.045 \\ 0.131 \\ 0.26 \\ 0.453 \\ 1 \end{bmatrix}$$

Figure 5: Matrix equation with substituted values

[1] *All matrixes where constructed using LaTeXit*

Using technology we can evaluate the system by inverting the Vandermonde matrix, see figure 6, in order to multiply the y values of the total income earned by the Vandermonde matrix consisting of the values of Brazil's population from lowest to highest income. Hence solving the system we obtain the coefficients (a_0, a_1, ...).

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.045 \\ 0.131 \\ 0.26 \\ 0.453 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0.2 & 0.04 & 0.008 & 0.0016 & 0.00032 \\ 1 & 0.4 & 0.16 & 0.064 & 0.0256 & 0.01024 \\ 1 & 0.6 & 0.36 & 0.216 & 0.1296 & 0.07776 \\ 1 & 0.8 & 0.64 & 0.512 & 0.4096 & 0.32768 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}^{-1}$$

Figure 6: Inverted Vandermonde matrix

Hence using technology the following values, see figure 7, are obtained for the coefficients in question. Their values rounded to 4 significant figures are also displayed in the figure, for the purpose of coherent presentation of results and simplification of further calculations.

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.3520833333 \\ -1.898958333 \\ 8.5625 \\ -12.52604167 \\ 6.510416667 \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0.3521 \\ -1.899 \\ 8.563 \\ -12.53 \\ 6.510 \end{bmatrix}$$

Figure 7: results and rounded results of matrix equation

Substituting the coefficients into the equation of the polynomial

$$P_n(x_n) = a_0 + a_1 x_n + a_2 x_n^2 + \ \dots \ + a_{n-1} x_n^{n-1} + a_n x_n^n$$

$$P(x) = 0 + 0.35120833333x - 1.8998958333 + 8.56325 - 12.52604167x^4 + 6.510416667x^5$$

Rounding the coefficients to 4 significant figures allows for sufficient accuracy while simplifying the continuing calculations.

$$P(x) = 6.510x^5 - 12.53x^4 + 8.563x^3 - 1.899x^2 + 0.3512x$$

Using Geogebra to graph the function and check if it passes through all points determined by Brazil's income distribution data:

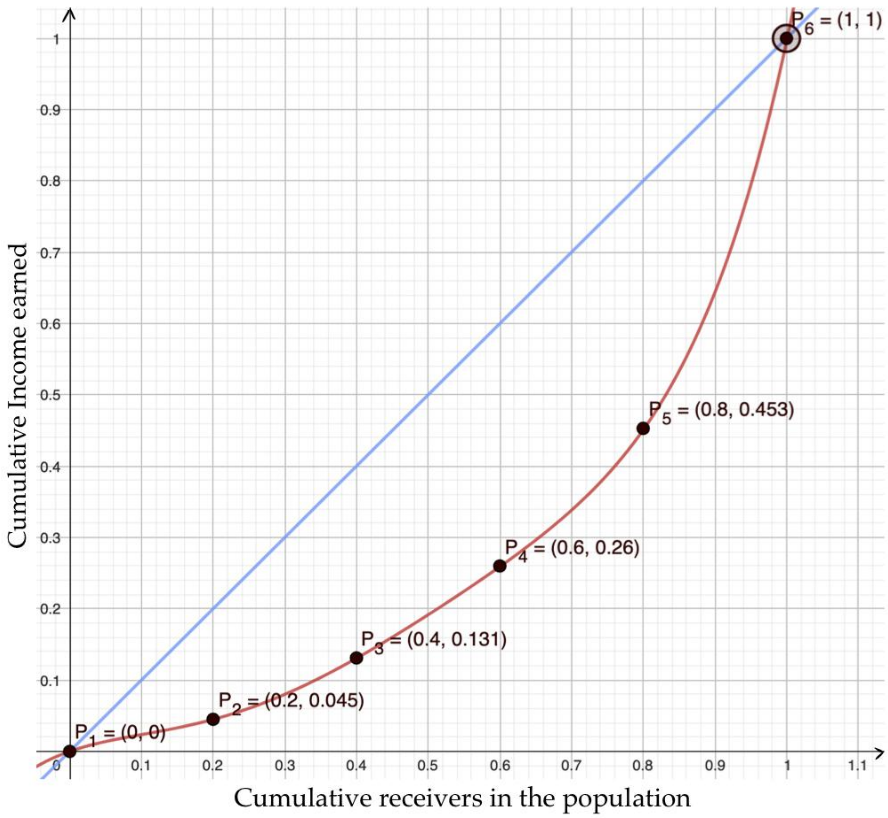

Figure 8: Plotted Lorenz curve from simultaneous equations using Geogebra

The Lorenz curve plots the cumulative percentages of total income earned in an economy on the y-axis against the cumulative number of receivers in the population of that economy from lowest to highest incomes on the x-axis.

Now integrating P(x) to find the area below the Lorenz curve (B)

$$Area\ B = \int_0^1 P(x) = \int_0^1 (6.510x^5 - 12.53x^4 + 8.563x^3 - 1.899x^2 + 0.3512x)dx$$

$$Area\ B = \left[\frac{6.510x^6}{6} - \frac{12..53x^5}{5} + \frac{8.563x^4}{4} - \frac{1.899x^3}{3} + \frac{0.3512x^2}{2}\right]_0^1$$

$$Area\ B = \left[\frac{6.510\times1^6}{6} - \frac{12.53\times1^5}{5} + \frac{8.563\times1^4}{4} - \frac{1.899\times1^3}{3} + \frac{0.3512\times1^2}{2} - (0)\right]$$

$$Area\ B = 0.26235\ units^2$$

Checking with calculator integration gives $\frac{5247}{20000} = 0.26235\, units^2$

From trapezium calculation we know Area that $A_{beneath\ equality\ line} = 0.5\, units^2$.

Now the Area A between equality line and Lorenz curve is

$A = 0.5 - 0.26235$

$A = 0.23765\, units^2$

Calculating the Gini coefficient using equation 1

$Gini\ coefficient = \frac{0.23765}{0.5}$

$Gini\ coefficient = 0.4753$

Rounded to two SF the Gini coefficient of 0.48 would indicate a 48% concentration of income distribution in Brazil which refers to a rather unequal distribution of income.
Compared to the value of 0.44 obtained with the trapezium sum method, this Gini coefficient value indicates a higher dispersion of income inequality because the area between the line of equality and Lorenz curve is greater than with the initially plotted points, thus showing a greater inequality. *(Hoang, Paul)*

4.3. Determining Brazil's Gini coefficient using Lagrange interpolation

As a challenge I decided to learn and apply the technique of Lagrange interpolation which does not form part of the mathematics course. Lagrange interpolation is a commonly used method in calculus to construct continuous functions based on discrete data. It aims to interpolate given data points exactly to the polynomial solution so that the values fit the resulting function.*(Lambers, Jim)* It is important to notice that the number of points of given data are always n+1 for a polynomial of degree n. Therefore, using data from Brazil's income distribution, we will use 6 interpolation points with the aim to construct the polynomial P(x) of degree 5, in order to integrate the resulting equation of the Lorenz curve to find the area beneath it. The general formula for finding the polynomial p(x) using Lagrange's method is equation 5.

Equation 5: $P_5(x) = \sum_{i=0}^{5} L_i(x)\, y_i$

Li(x) are the Lagrange terms and represent an n degree polynomial with n roots that passes through (n+1) given data points. Since we have six data points given for Brazil's Lorenz curve n=5, we will obtain a polynomial of degree 5. *(Shao, Jun)*

The Lagrange terms L can be determined with equation 6 using product notation *(Ford, Whit)*. While sigma notation Σ refers to the sum of a number of n terms, whereas the product notation with the greek letter capital pi Π refers to repeated multiplication as it represents the product of a number of factors. *(Mechtutor com)*

It is a symbol that expresses this mathematical operation in a compact notation, accompanied by a multiplication index j below the Π symbol that indicates the number of terms n, the upper boundary, that we multiply to the right of Π. *(Adam Hrankowski)*

Equation 6: $L_i(x) = \prod_{j=1,\, j\neq i}^{n} \frac{x-x_j}{x_i-x_j}$; To avoid a 0 in a denominator j can never be equal to i

(Lambers, Jim)

Combining these two equations and considering the polynomial of degree 5 would give:

Equation 7: $P_5(x) = \sum_{i=0}^{5} \left(\prod_{j=1,\, j\neq i}^{5} \frac{x-x_j}{x_i-x_j} \right) y_i$

The polynomial equation we want to find for Brazil's Lorenz curve can therefore be expressed as

$$P_5(x) = \sum_{i=0}^{5} L_i(x) y_i = L_1 \times y_0 + L_2 \times y_1 + L_3 \times y_2 + L_4 \times y_3 + L_5 \times y_4 + L_6 \times y_5$$

We are given 6 points (x_n, y_n) from Brazil's data and are finding a 5th order polynomial.

Table 4: Brazil's data for Lagrange interpolation

n	x_n	y_n
0	0	0
1	0.2	0.045
2	0.4	0.131
3	0.6	0.26
4	0.8	0.453
5	1	1

Substituting the y values into the equation we get:

$$P(x) = L_1 \times 0 + L_2 \times 0.045 + L_3 \times 0.131 + L_4 \times 0.26 + L_5 \times 0.453 + L_6 \times 1$$

Using equation 7 for each specific order we can construct each Lagrange polynomial as follows:

$$L_0(x) = \frac{(x-x_1)}{(x_0-x_1)} \times \frac{(x-x_2)}{(x_0-x_2)} \times \frac{(x-x_3)}{(x_0-x_3)} \times \frac{(x-x_4)}{(x_0-x_4)} \times \frac{(x-x_5)}{(x_0-x_5)}$$

$$L_1(x) = \frac{(x-x_0)}{(x_1-x_0)} \times \frac{(x-x_2)}{(x_1-x_2)} \times \frac{(x-x_3)}{(x_1-x_3)} \times \frac{(x-x_4)}{(x_1-x_4)} \times \frac{(x-x_5)}{(x_1-x_5)}$$

$$L_2(x) = \frac{(x-x_0)}{(x_2-x_0)} \times \frac{(x-x_1)}{(x_2-x_1)} \times \frac{(x-x_3)}{(x_2-x_3)} \times \frac{(x-x_4)}{(x_2-x_4)} \times \frac{(x-x_5)}{(x_2-x_5)}$$

$$L_3(x) = \frac{(x-x_0)}{(x_3-x_0)} \times \frac{(x-x_1)}{(x_3-x_1)} \times \frac{(x-x_2)}{(x_3-x_2)} \times \frac{(x-x_4)}{(x_3-x_4)} \times \frac{(x-x_5)}{(x_3-x_5)}$$

$$L_4(x) = \frac{(x-x_0)}{(x_4-x_0)} \times \frac{(x-x_1)}{(x_4-x_1)} \times \frac{(x-x_2)}{(x_4-x_2)} \times \frac{(x-x_3)}{(x_4-x_3)} \times \frac{(x-x_5)}{(x_4-x_5)}$$

$$L_5(x) = \frac{(x-x_0)}{(x_5-x_0)} \times \frac{(x-x_1)}{(x_5-x_1)} \times \frac{(x-x_2)}{(x_5-x_2)} \times \frac{(x-x_3)}{(x_5-x_3)} \times \frac{(x-x_4)}{(x_5-x_4)}$$

Substituting the x values from Brazil's data gives following equations

$$L_0(x) = \frac{(x-0.2)}{(0-0.2)} \times \frac{(x-0.4)}{(0-0.4)} \times \frac{(x-0.6)}{(0-0.6)} \times \frac{(x-0.8)}{(0-0.8)} \times \frac{(x-1)}{(0-1)}$$

$$L_1(x) = \frac{(x-0)}{(0.2-0)} \times \frac{(x-0.4)}{(0.2-0.4)} \times \frac{(x-0.6)}{(0.2-0.6)} \times \frac{(x-0.8)}{(0.2-0.8)} \times \frac{(x-1)}{(0.2-1)}$$

$$L_2(x) = \frac{(x-0)}{(0.4-0)} \times \frac{(x-0.2)}{(0.4-0.2)} \times \frac{(x-0.6)}{(0.4-0.6)} \times \frac{(x-0.8)}{(0.4-0.8)} \times \frac{(x-1)}{(0.4-1)}$$

$$L_3(x) = \frac{(x-0)}{(0.6-0)} \times \frac{(x-0.2)}{(0.6-0.2)} \times \frac{(x-0.4)}{(0.6-0.4)} \times \frac{(x-0.8)}{(0.6-0.8)} \times \frac{(x-1)}{(0.6-1)}$$

$$L_4(x) = \frac{(x-0)}{(0.8-0)} \times \frac{(x-0.2)}{(0.8-0.2)} \times \frac{(x-0.4)}{(0.8-0.4)} \times \frac{(x-0.6)}{(0.8-0.6)} \times \frac{(x-1)}{(0.8-1)}$$

$$L_5(x) = \frac{(x-0)}{(1-0)} \times \frac{(x-0.2)}{(1-0.2)} \times \frac{(x-0.4)}{(1-0.4)} \times \frac{(x-0.6)}{(1-0.6)} \times \frac{(x-0.8)}{(1-0.8)}$$

The individual terms of $P_n(x)$ considering equation 6 will then be as follows:

$Term\,0 = 0$

$Term\,1 = 5.859375 \times (x) \times (x - 0.4) \times (x - 0.6) \times (x - 0.8) \times (x - 1)$

$Term\,2 =- 34.11458333 \times (x) \times (x - 0.2) \times (x - 0.6) \times (x - 0.8) \times (x - 1)$

$Term\,3 = 67.70833333 \times (x) \times (x - 0.2) \times (x - 0.4) \times (x - 0.8) \times (x - 1)$

$Term\,4 =- 58,984375 \times (x) \times (x - 0.2) \times (x - 0.4) \times (x - 0.6) \times (x - 1)$

$Term\,5 = 26.04166667 \times (x) \times (x - 0.2) \times (x - 0.4) \times (x - 0.6) \times (x - 8)$

Simplifying these terms using technology to avoid human error and summing them together gives the equation for the Lorenz curve as:

$$P_5(x) = 6.51041667x^5 - 12.5260416x^4 + 8.5624995x^3 - 1.8989582x^2 + 0.35208334x$$

Simplifying the equation to an equal number of significant figures by rounding the coefficients to 4 significant figures has the purpose to simplify further calculations while maintaining accuracy.

$$P_5(x) = 6.510x^5 - 12.53x^4 + 8.562x^3 - 1.899x^2 + 0.3521x$$

Using Geogebra to graph the function and check if it passes through all points determined by Brazil's income distribution data:

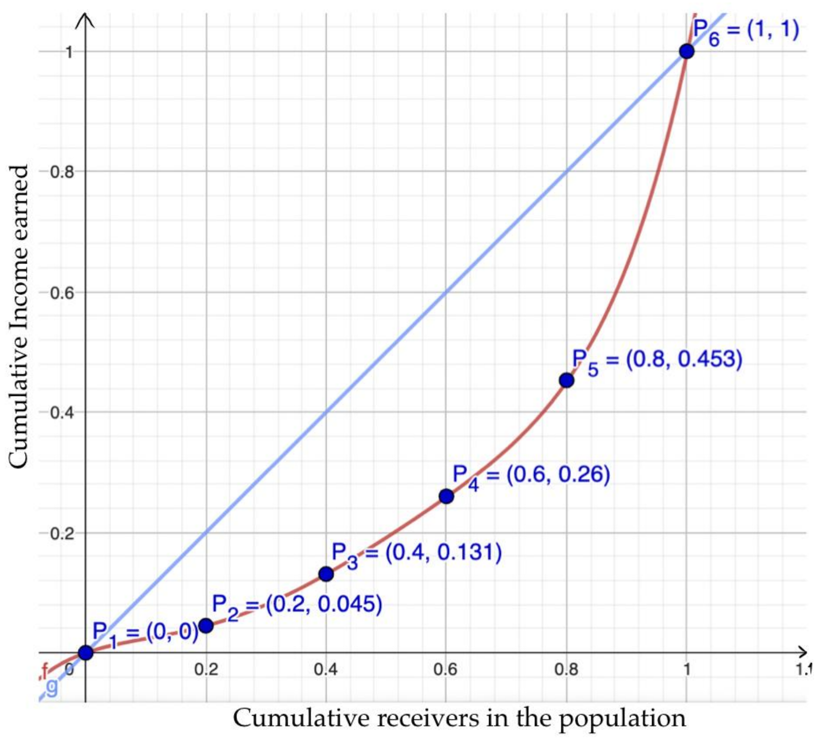

Figure 9: Plotted Lorenz curve from Lagrange interpolation using Geogebra

Now to calculate the Gini coefficient the area B beneath the red Lorenz curve can be found by integrating P(x):

$$Area\ B = \int_0^1 P(x) = \int_0^1 (6.510x^5 - 12.53x^4 + 8.562x^3 - 1.899x^2 + 0.3521x\ )dx$$

$$Area\ B\ = \left[\frac{6.510x^6}{6} - \frac{12..53x^5}{5} + \frac{8.562x^4}{4} - \frac{1.899x^3}{3} + \frac{0.3521x^2}{2}\right]_0^1$$

$$Area\ B\ = \left[\frac{6.510\times1^6}{6} - \frac{12.53\times1^5}{5} + \frac{8.562\times1^4}{4} - \frac{1.899\times1^3}{3} + \frac{0.3521\times1^2}{2} - (0)\right]$$

$$Area\ B\ = 0.26255\ units^2$$

Checking with calculator integration gives $\frac{5251}{20000} = 0.26255\ units^2$

From trapezium calculation we know Area that $A_{beneath\ equality\ line} = 0.5\ units^2$.

Now the Area A between equality line and Lorenz curve is:

$$A\ = 0.5\ -\ 0.26255$$

$$A\ = 0.23745\ units^2$$

Calculating the Gini coefficient using equation 1

$$Gini\ coefficient\ = \frac{0.23745}{0.5}$$

$$Gini\ coefficient\ = 0.4749$$

Rounded to two significant figures the Gini coefficient of 0.47 would indicate a 47% concentration of income distribution in Brazil which refers to a rather unequal distribution of income. This Gini coefficient is similar to the previously obtained value using a system of equations and thus indicates a higher inequality in income distribution. Compared to the first value of 0.44, this Gini coefficient demonstrates a “worse” situation of inequality in Brazil because the area between line of equality and Lorenz curve is greater *(Hoang, Paul).*

5.1 Reflection, evaluation and conclusion

The most recent data set I found in order to construct the Lorenz curve is from 2020 and thus has not taken into account recent global economic occurrences like the medium term effects and shortages due to the covid pandemic as well as the geopolitical crisis in the Ukraine. According to *Our World data* as well as *The World Bank,* Brazil’s Gini coefficient was 0.49 in 2020 *(Our World in Data)(World Bank).* Using this value error calculations can be performed for the three mathematical approaches explored in determining the Gini coefficient.

Approximation error calculations:

Equation 8: $\delta = \left|\frac{v_O - v_A}{v_A}\right| \times 100\%$

δ is the percentage error, v_O is the observed value and v_A the actual value expected.

For trapezium method v_O=0.4444 and percentage error 9.306% ≈ 9.3%.
For simultaneous equations v_O=0.4753 and percentage error 3%.
For Lagrange interpolation v_O=0.4749 and percentage error 3.082% ≈ 3.1% .

The aim of this mathematical exploration was to construct the Lorenz curve and use three different mathematical methods to calculate Brazil's Gini coefficient in 2020. The methods explored were the Trapezium method and two different interpolation methods, using simultaneous equations and Lagrange polynomials. The Trapezium method is by far the easiest mathematical approximation of the area beneath a curve as it was easy to understand and apply to Brazil's Lorenz curve. By summing up the trapezia beneath the curve no integration was necessary during this procedure. However, this method is a form of linear approximation and therefore its result was expected to be less accurate than for the polynomial interpolation methods due to the continuous curved shape of the Lorenz curve which, constructing the curve from 6 given data points, was not achieved. The error calculations performed justify that since the trapezium method's result for the Gini coefficient has with a percentage error of 9.3% the highest deviation from the actual value for Brazil's Gini coefficient. The simultaneous equations and interpolation method were more accurate as they considered the curved shape of the Lorenz curve by determining the polynomial function. However, they were much more complicated in application particularly due to the 6 data points given. Both approaches gave nearly the same value with approximately 3% percentage error. The simultaneous equations proved to be the initially more difficult to understand approach as the use of matrices may seem overwhelming at times but easier to actually apply once understood since Lagrange polynomials are difficult to keep track of numbers with a large amount of decimals. This allows a greater scope for rounding errors and miscalculations. Furthermore Lagrange requires that for each point calculations have to be repeated making it a very extensive approach. Using simultaneous equations was the most accurate method used with a percentage error of 3%. However, solving requires a large amount of columns and rows for which it is more suitable to use the calculator for evaluation than solving the system of equations by hand. Using a large matrix leads to accumulation of round-off errors as well. Therefore for both interpolation methods the drawbacks are the rounding to fewer decimal places and mathematical work involving a large system of equations which may have been the reason as to why the answers differed from the actual published value of Brazil's Gini coefficient. A possible extension for this exploration could be the row reduction method in solving the simultaneous equations instead of using technology as well as research and application of further interpolation methods for polynomials, for instance the Newton interpolation formula.

To conclude, the mathematical methods to determine the Gini coefficient explored in this IA vary in accuracy and difficulty. The Trapezium method was the easiest method used but had the largest deviation and error in Gini coefficient, whereas the interpolation methods of simultaneous equations and Lagrange gave a similar result with a much lower error percentage than the trapezium method. They did however take longer to understand and apply, at times repeated attempts were necessary to understand errors made. This exploration provided insight into the mathematical analysis and determination of the Gini coefficient which is an important economic indicator in our modern world that describes the distribution of income or wealth in a given country or economy. In particular the two interpolation methods proved to give the most accurate results for the real life example of Brazil's income distribution when compared to the actual value of the Gini coefficient published by World Data. The results obtained appear valid considering the drawbacks and errors of calculations identified and demonstrated that both interpolation methods are a reliable and effective approximation of the Gini coefficient, whereas the trapezium method proved to be more efficient rather than accurate.

6. Works cited

Figure 1: The Gini coefficient, Hoang, Paul, et al. "Measuring Economic Inequality." *Economics for the IB Diploma Programme*, Hodder Education, London, England, 2020, pp. 330–332.

Figure 2 - 9: Created by Internal assessment author

Adam Hrankowski, ADHD. "How to Use Π Product Notation." Medium, MathAdam, 15 Oct. 2020, https://medium.com/mathadam/how-to-use-%CF%80-product-notation-ae4a4c257471.

CFI Team. "Gini Coefficient." *Corporate Finance Institute*, CFI Education Inc., 6 Dec. 2022, https://corporatefinanceinstitute.com/resources/economics/gini-coefficient/.

Ford, Whit. “Pi Notation (Product Notation).” *MathMaine*, Wordpress.com, 24 Sept. 2021, https://mathmaine.com/2018/03/04/pi-notation/.

Hayes, Adam. “Gini Index Explained and Gini Co-Efficients around the World.” *Investopedia*, Investopedia, 13 Jan. 2023, https://www.investopedia.com/terms/g/gini-index.asp.

Hoang, Paul, et al. “Measuring Economic Inequality.” *Economics for the IB Diploma Programme*, Hodder Education, London, England, 2020, pp. 330–332.

Lambers, Jim. “Lagrange Interpolation - USM.” *Math.usm*, School Of Mathematical Sciences , 2010, https://www.math.usm.edu/lambers/mat772/fall10/lecture5.pdf.

Mechtutor com, director. *Lagrange Interpolation Method: Algorithm, Computation and Plot | Numerical Computing with Python*. *YouTube*, Mechtutor Come, 12 Feb. 2020, https://www.youtube.com/watch?v=dTGqOj1NZwY. Accessed 20 Feb. 2023.

Our World in Data. “Income Inequality: Gini Coefficient, 2020.” *Our World in Data*, University of Oxford, 2020, https://ourworldindata.org/grapher/economic-inequality-gini-index?tab=chart&time=latest&country=~BRA.

Ramzai, Juhi. “Clearly Explained: Gini Coefficient and Lorenz Curve.” *Medium*, Towards Data Science, 27 Apr. 2020, https://towardsdatascience.com/clearly-explained-gini-coefficient-and-lorenz-curve-fe6f5dcdc07.

Shao, Jun. “Interpolation Methods - University of Wisconsin–Madison.” Mathematical Statistics, University of Wisconsin Madison, 2018, https://pages.cs.wisc.edu/~sifakis/courses/cs412-s13/lecture_notes/CS412_5_Feb_2013.pdf

The Investopedia Team. “Lorenz Curve.” Edited by Eric Estevez, *Investopedia*, Investopedia, 19 Dec. 2022, https://www.investopedia.com/terms/l/lorenz-curve.asp.

The Math Guy, director. *Interpolation - Vandermonde Matrix*. *YouTube*, The Math Guy, 11 Oct. 2017, https://www.youtube.com/watch?v=G6lVSD0Jci0. Accessed 20 Feb. 2023.

World Bank. “Gini Index - Brazil.” *Data*, World Bank, 2020, https://data.worldbank.org/indicator/SI.POV.GINI?locations=BR.

World Bank. “Income Share Held by Lowest 20% - Brazil.” *Data*, The World Bank, 2020, https://data.worldbank.org/indicator/SI.DST.FRST.20?locations=BR.

World Bank. “Income Share Held by Lowest 20% - Germany.” *Data*, The World Bank, 2018, https://data.worldbank.org/indicator/SI.DST.FRST.20?end=2018&locations=DE&start=1991&view=chart.